Image Segmentation Using Thresholding Techniques in OpenCV

Developed By

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Aim

To segment an image using Global Thresholding, Adaptive Thresholding, and Otsu's Thresholding techniques using Python and OpenCV.

The program performs the following operations:

- Global Thresholding
- Adaptive Thresholding
- Otsu's Thresholding

Software Used

- Anaconda – Python 3.7
- Jupyter Notebook / VS Code
- OpenCV (cv2)
- NumPy
- Matplotlib

Algorithm

Step 1:

Import the required libraries: OpenCV, NumPy, and Matplotlib.

Step 2:

Load the input image using OpenCV.

Step 3:

Convert the input image into grayscale format.

Step 4: Global Thresholding

- Select a fixed threshold value.
- Apply thresholding to separate foreground and background pixels.
- Display the thresholded image.

Step 5: Adaptive Thresholding

- Compute threshold values for small regions of the image.
- Apply Adaptive Mean Thresholding.
- Apply Adaptive Gaussian Thresholding.
- Display the segmented images.

Step 6: Otsu's Thresholding

- Automatically determine the optimal threshold value.
- Apply Otsu's thresholding technique.

- Display the segmented image.

Step 7:

Compare the results obtained from Global, Adaptive, and Otsu's thresholding methods.

Program :

Original Grayscale Image

```
import cv2
import matplotlib.pyplot as plt
img = cv2.imread("lego.jpg")
plt.imshow(cv2.cvtColor(img, cv2.COLOR_BGR2RGB))
plt.title("Original Image")
plt.axis("off")
plt.show()
```

Original Image

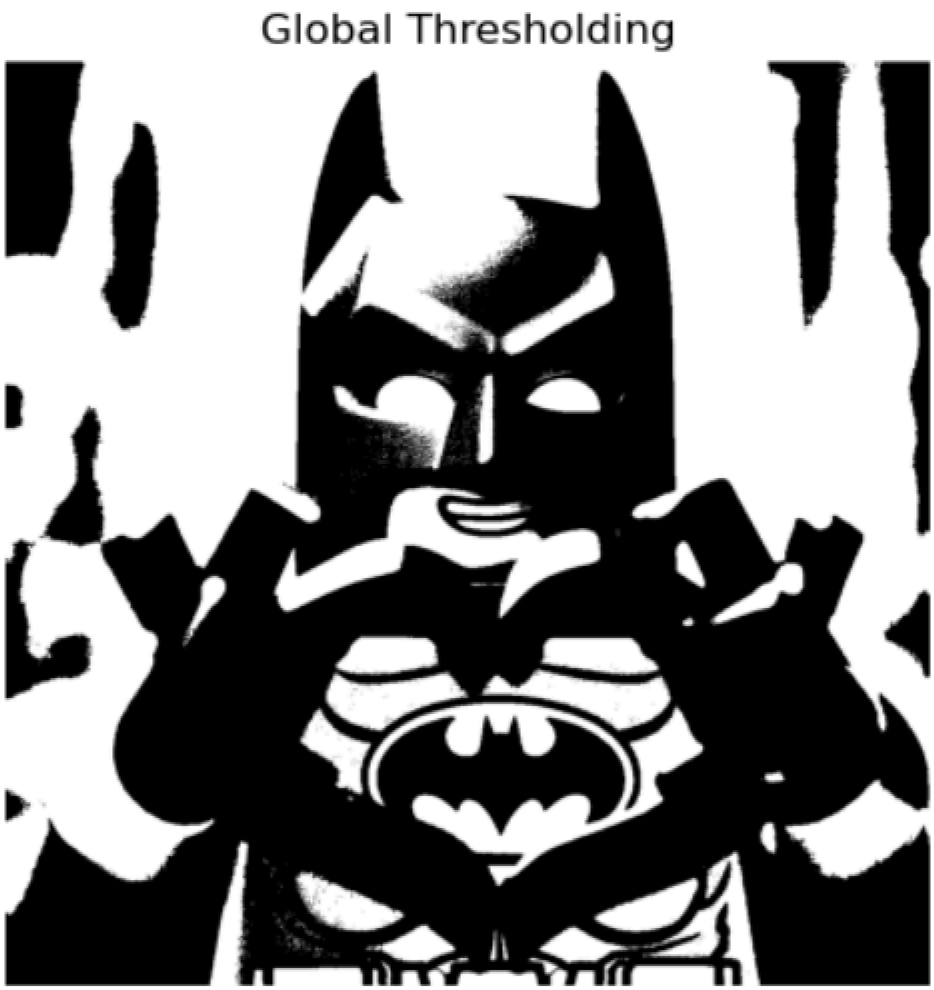
Global Thresholding

```
import cv2
import matplotlib.pyplot as plt
img = cv2.imread("lego.jpg", cv2.IMREAD_GRAYSCALE)
plt.imshow(img, cmap="gray")
plt.title("Original Grayscale Image")
plt.axis("off")
plt.show()
```

Adaptive Thresholding

```
import cv2
import matplotlib.pyplot as plt
img = cv2.imread("lego.jpg", cv2.IMREAD_GRAYSCALE)
_, result = cv2.threshold(img, 127, 255, cv2.THRESH_BINARY)
plt.imshow(result, cmap="gray")
plt.title("Global Thresholding")
```

```
plt.axis("off")
plt.show()
```



Otsu's Thresholding

```
import cv2
import matplotlib.pyplot as plt
img = cv2.imread("lego.jpg", cv2.IMREAD_GRAYSCALE)
_, result = cv2.threshold(
    img, 0, 255,
    cv2.THRESH_BINARY + cv2.THRESH_OTSU
)
plt.imshow(result, cmap="gray")
plt.title("Otsu's Thresholding")
plt.axis("off")
plt.show()
```

Result

Thus, image segmentation is successfully performed using **Global Thresholding, Adaptive Thresholding, and Otsu's Thresholding** techniques in OpenCV.